Kavimalar D 241801118 exp 4

```
!pip install nltk -q

import pandas as pd
import numpy as np
import re
import string
import nltk

from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')

df = pd.read_csv("/content/Reviews.csv")

print("Shape:", df.shape)
display(df.head())
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
Shape: (106337, 10)
```

	Id	ProductId	UserId	ProfileName	HelpfulnessNumerator	HelpfulnessDenominator	Score	Time	Sum
0	1	B001E4KFG0	A3SGXH7AUHU8GW	delmartian	1.0	1.0	5.0	1.303862e+09	Q Dog
1	2	B00813GRG4	A1D87F6ZCVE5NK	dll pa	0.0	0.0	1.0	1.346976e+09	N Adve

```
df = df[['Text']].dropna()
df = df.drop_duplicates()
df = df.head(10000).reset_index(drop=True)

print("Number of reviews:", len(df))
display(df.head())
```

```
Number of reviews: 10000
```

	Text
0	I have bought several of the Vitality canned d...
1	Product arrived labeled as Jumbo Salted Peanut...
2	This is a confection that has been around a fe...
3	If you are looking for the secret ingredient i...
4	Great taffy at a great price. There was a wid...

```
stop_words = set(stopwords.words('english'))

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-zA-Z\s]', '', text)
    tokens = word_tokenize(text)
    tokens = [word for word in tokens if word not in stop_words]
    return ' '.join(tokens)

df['Cleaned_Text'] = df['Text'].apply(preprocess)

display(df[['Text', 'Cleaned_Text']].head())
```

	Text	Cleaned_Text
0	I have bought several of the Vitality canned d...	bought several vitality canned dog food produc...
1	Product arrived labeled as Jumbo Salted Peanut...	product arrived labeled jumbo salted peanutsth...
2	This is a confection that has been around a fe...	confection around centuries light pillowy citr...
3	If you are looking for the secret ingredient i...	looking secret ingredient robitussin believe f...
4	Great taffy at a great price. There was a wid...	great taffy great price wide assortment yummy ...

```
vectorizer = TfidfVectorizer()

tfidf_matrix = vectorizer.fit_transform(df['Cleaned_Text'])

print("TF-IDF Matrix Shape:", tfidf_matrix.shape)
print("Vocabulary Size:", len(vectorizer.vocabulary_))
```

```
TF-IDF Matrix Shape: (10000, 24267)
Vocabulary Size: 24267
```

```
def retrieve_reviews(query):
    query_cleaned = preprocess(query)

    query_vector = vectorizer.transform([query_cleaned])

    similarity = cosine_similarity(query_vector, tfidf_matrix).flatten()

    top_indices = similarity.argsort()[-5:][::-1]

    results = df.iloc[top_indices].copy()
    results['Similarity Score'] = similarity[top_indices]

    return results[['Similarity Score', 'Text', 'Cleaned_Text']]
```

```
queries = [
    "great product with fast shipping",
    "disappointed",
    "excellent taste",
    "poor packaging"
]

for query in queries:
    print("\n" + "="*80)
    print("QUERY:", query)
    print("="*80)
    display(retrieve_reviews(query))
```

```
================================================================================
QUERY: great product with fast shipping
================================================================================
```

	Similarity Score	Text	Cleaned_Text
5210	0.498952	Enjoyed the product and they also provided ver...	enjoyed product also provided fast shipping im...
7992	0.441688	The tea is good and fresh. We enjoy it. The sh...	tea good fresh enjoy shipping fast cost reason...
6011	0.401147	The energy drink is a great product. The shipp...	energy drink great product shipping price craz...
7048	0.392853	My daughter lives in Hawaii and sent me some g...	daughter lives hawaii sent great coffee keurig...
9391	0.388207	Fast shipping, items were packaged nicely and ...	fast shipping items packaged nicely described ...

```
================================================================================
QUERY: disappointed
================================================================================
```

	Similarity Score	Text	Cleaned_Text
3140	0.487114	I am a bit disappointed. The flavor was not w...	bit disappointed flavor wanted expected
4365	0.359892	The product is very good. Way too expensive an...	product good way expensive almost box get panc...
6523	0.321561	Disappointed. The big boxes had a very differ...	disappointed big boxes different flavor smalle...
3322	0.312904	Just plain nasty!!! This item tasted like card...	plain nasty item tasted like cardboard watered...
6482	0.312098	I was a bit disappointed in these. They crumb...	bit disappointed crumble easily unusual textur...

```
================================================================================
QUERY: excellent taste
================================================================================
```

	Similarity Score	Text	Cleaned_Text
1740	0.453237	The Popchips were excellent - had excellent fl...	popchips excellent excellent flavor low calori...
7298	0.407300	Excellent product. Exactly what they said it w...	excellent product exactly said would superior ...
3396	0.386284	We recieved the product on time and in excelle...	recieved product time excellent shape tastes g...
9471	0.383315	Title says it all, a Excellent quality product...	title says excellent quality product excellent...
2278	0.357031	This is an excellent tea, very flavorful - the...	excellent tea flavorful cinnamon fresh powerfu...

```
================================================================================
QUERY: poor packaging
================================================================================
```

	Similarity Score	Text	Cleaned_Text
6162	0.324483	I was delighted to find this product online, b...	delighted find product online always find stor...
4595	0.309008	This review is for the packaging and not the a...	review packaging actual mix since mix subjecti...
7553	0.306153	The soup is rich and tasty, but DON'T buy it f...	soup rich tasty dont buy amazon packaging poor...
2002	0.298929	Didnt really enjoy these after wanting to try ...	didnt really enjoy wanting try something aunt ...
9663	0.259143	was really dissappointed in product... too sal...	really dissappointed product salty poor flavor...

```
for query in queries:
    results = retrieve_reviews(query)
```

```
    print("\nQuery:", query)
    print("Average Similarity Score:",
          round(results['Similarity Score'].mean(), 4))

    print("Top Review:")
    print(results.iloc[0]['Text'][:300])
    print()
```

```
Query: great product with fast shipping
Average Similarity Score: 0.4246
Top Review:
Enjoyed the product and they also provided very fast shipping.  I'm about out and need to order more.


Query: disappointed
Average Similarity Score: 0.3587
Top Review:
I am a bit disappointed.  The flavor was not what I wanted or expected.
```